

SMART INDIA HACKATHON 2026



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HACKATHON
2026

SIH 2026 • PS SIH26173 • ISRO



iTantra

iTantra

Voice that travels as **meaning**

Speech in, a 234-byte signed packet across a phone-to-phone mesh, speech out. No tower, no internet, 10 Indian languages, on 3 GB phones.

PROBLEM STATEMENT TITLE

iTantra: Indian Multilingual TTS & STT Aided Neural Transceiver Radio Access for low bitrate links

PROBLEM STATEMENT ID

SIH26173

THEME

Miscellaneous

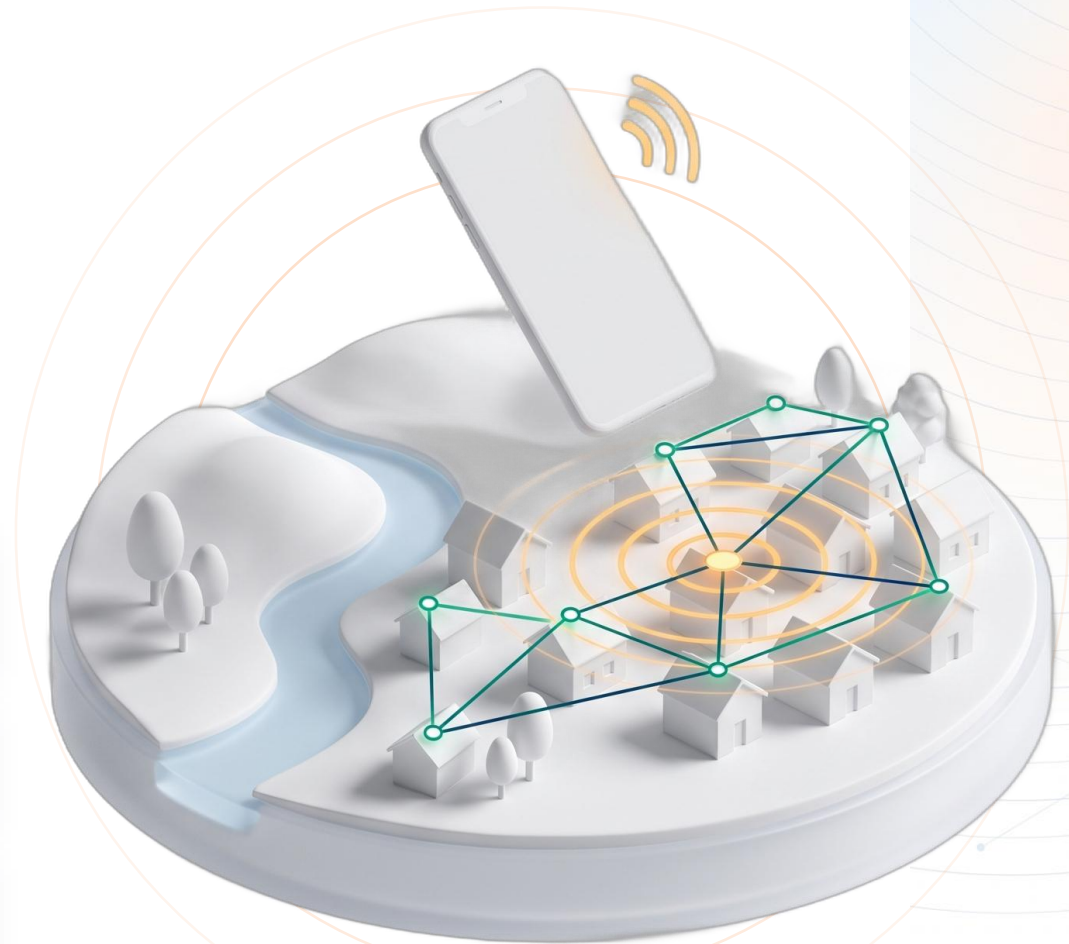
PS CATEGORY

Software

TEAM ID

TEAM NAME (REGISTERED ON PORTAL)

Hexabits



 100% offline

 234 B / sentence

 10 languages

 Alert mode

iTantra: Send the Meaning, Not the Audio

Hexabits

01 Proposed Solution (Describe your Idea/Solution/Prototype)

Detailed explanation: speech is recognised on the phone, sent as a few hundred bytes of signed text over a phone-to-phone mesh, and spoken again on the receiving phone.



1 Speak + on-device STT

VAD finds the pause, IndicConformer writes it



2 234 B packet hops the mesh

BLE + Wi-Fi Direct relays, up to 7 hops, no tower



3 On-device TTS speaks it

Voice note; ALERTs ring at max volume

How it addresses the problem: bytes to send one 3.5 s sentence

Raw voice, 64 kbps  28,000 B

Opus codec, 16 kbps  7,000 B

iTantra packet  234 B  120 × smaller

 No tower, internet or server

 Heard by people who can't read

 Runs on 3 GB phones

Innovation and uniqueness



Semantic radio

Sends meaning + prosody, not the waveform



Alert-first mesh

Floods all hops; loud until acknowledged



Tamper-proof

Ed25519-signed; private messages encrypted



100% on-device

No cloud API; packs download once

02 Technologies to be used

APP & UI

Flutter (Dart)

Kotlin (Android)

Pigeon bridge

Riverpod

ON-DEVICE AI

sherpa-onnx

IndicConformer int8 (STT)

MMS VITS (TTS)

Silero VAD

MESH & SECURITY

BLE GATT mesh

Wi-Fi Direct

Ed25519

X25519 + ChaCha20

MODEL TOOLING

Python

ONNX int8

Google Colab

Hugging Face packs

03 Methodology and process for implementation (flow)



04 Working prototype (Android)



644 ms

STT model load

236 ms

Hindi sentence decode

654 ms

TTS voice load

10 / 10

languages verified

App screen redrawn from the running build; timings measured on OnePlus CPH2717, Android 16.

05 Analysis of the feasibility of the idea
 **10 / 10**

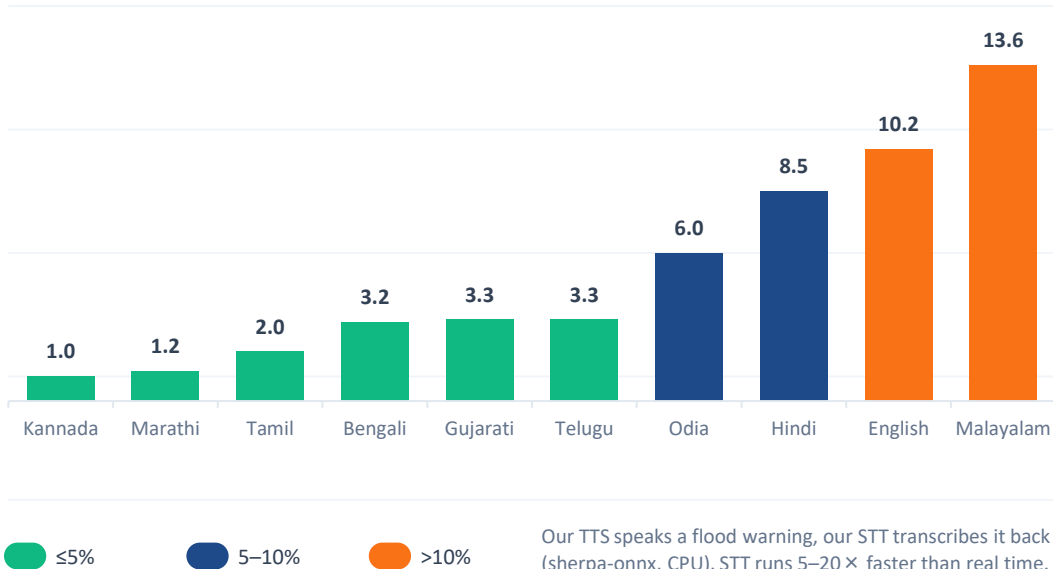
languages running fully offline

 **~0.3 GB**

per language: speech + voice packs

 **<1.1 GB**

peak RAM (est.) fits 3 GB phones

Round-trip test: character error % per language (lower is better)


Our TTS speaks a flood warning, our STT transcribes it back (sherpa-onnx, CPU). STT runs 5–20 × faster than real time.

06 Potential challenges and risks → Strategies to overcome them
 Short range per hop

~60–100 m per Bluetooth hop


 7-hop relay, BLE Coded PHY where supported, store-and-forward by moving phones

 Hard speech

Noise, soft voices, Malayalam / Odia


 Gain control + tuned VAD, distress hotword list, text shown with voice, real-speech test set

 Low-end phones

RAM, battery drain, background kills


 int8 models, one language in RAM, duty-cycled scanning, foreground service

 Fake alerts, misuse

Spoofing; mesh apps flagged in India (Jul 2026)


 Ed25519-signed packets, verified NDMA / ISRO key, positioned as authenticated alert tool

 Voice licence

MMS-TTS is non-commercial


 Swap to CC-BY voices trained on SYSPIN / IndicVoices-R before deployment



85%

of Indians speak one of our 10
languages as mother tongue

Census 2011



~26%

of Indians aged 7+ cannot read: voice
still reaches them

Census 2011



0

towers, internet, SIM cards or
servers needed
fully offline

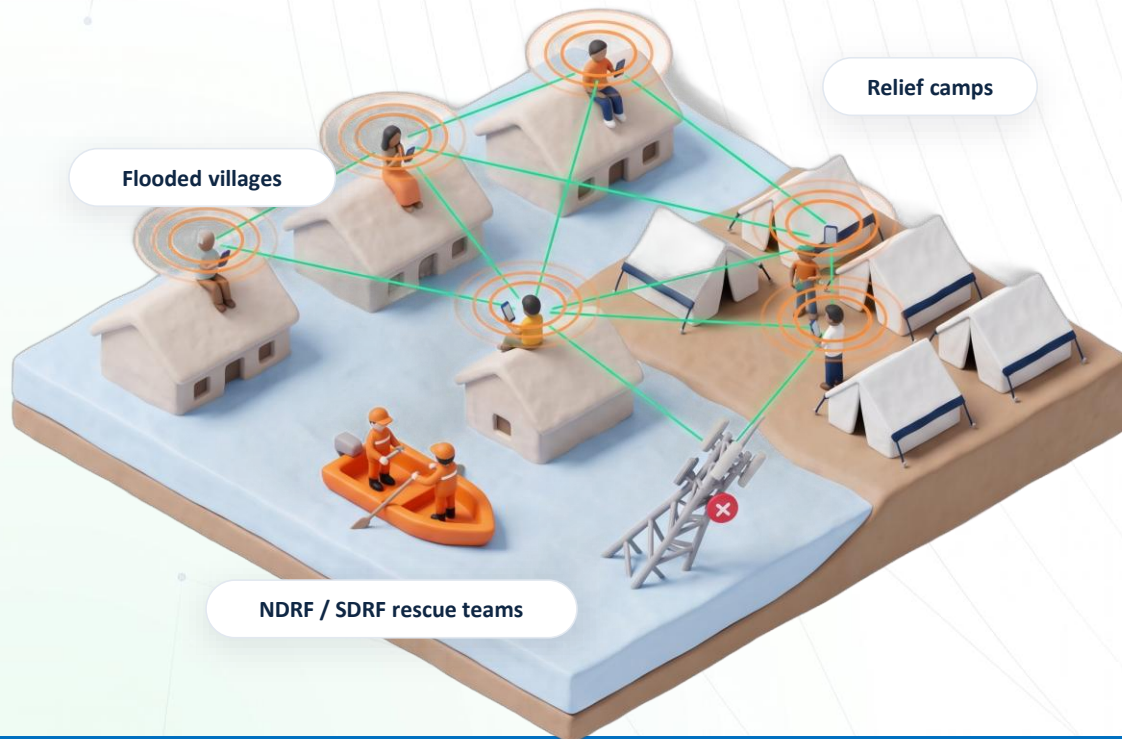


₹0

data cost per message, on phones
people already own
no data plan

07

Potential impact on the target audience



08

Benefits of the solution



Social

Voice in 10 languages for non-literate, elderly
and visually impaired people; no typing.



Economic

Runs on existing low-cost phones; zero
infrastructure, data or SMS cost; open models.



Environmental

No diesel towers or generators; low-power
Bluetooth saves battery in long outages.



Strategic (ISRO)

Byte-sized frames suit NavIC messaging and LoRa
links; data stays on Indian devices.



NEXT "heard by N phones" receipts • SMS gateway on signal • voice-preserving prosody •
10-phone field trial

09

Details / Links of the reference and research work



Foundation models

AI4Bharat IndicConformer ASR

MIT

github.com/AI4Bharat/IndicConformerASR

sherpa-onnx offline runtime

Apache-2.0

github.com/k2-fsa/sherpa-onnx

Meta MMS-TTS voices

VITS

huggingface.co/facebook/mms-tts

Silero VAD

MIT

github.com/snakers4/silero-vad

IndicConformer sherpa int8 exports

HF

huggingface.co/.../indicconformer-sherpa-onnx

Academic literature

IndicVoices: Indian speech dataset

ACL 2024

arxiv.org/abs/2403.01926

STCTS: speech at 80 bps (text + prosody)

arXiv 2025

arxiv.org/abs/2512.00451

IndicVoices-R: Indian TTS corpus

NeurIPS 2024

github.com/AI4Bharat/IndicVoices-R

SYSPIN open Indian TTS dataset

IISc

vaani.iisc.ac.in/dataset/syspindataset

BitChat mesh protocol whitepaper

Open

github.com/permissionlesstech/bitchat

Field precedents & data

FireChat mesh, Hong Kong 2014

500k installs

en.wikipedia.org/wiki/FireChat

BitChat in Nepal 2025

48,781 / day

forbes.com/sites/digital-assets

Bridgefy mesh adds E2E encryption

TechCrunch 2020

techcrunch.com/2020/11/02

Census of India 2011

Language, literacy

censusindia.gov.in

NDMA disaster management

Govt. of India

ndma.gov.in

Built, verified and open source

Android prototype

20 language packs

10/10 round-trip

Scan to see the code and download the offline model packs.



App source code

github.com/nileshpatil6/SIH-Hexabits

Offline model packs

huggingface.co/datasets/Mr66/itantra-packs